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Metrology method

Abstract

A method includes placing a wafer on a rotation mechanism of a metrology device; illuminating, by using a light source of the metrology device, the wafer by an X-ray; rotating, by using the rotation mechanism, the wafer while illuminating the wafer by the X-ray; detecting, by using an image sensor of the metrology device, a transmission portion of the X-ray passing through the wafer while rotating the wafer; and obtaining, by using a processor of the metrology device, a top width and a bottom width of a structure over the wafer based on the transmission portion of the X-ray with different rotating angles of the rotation mechanism.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6272198	12/2000	Shimura	N/A	N/A
7139365	12/2005	Janik	N/A	N/A
7303132	12/2006	Knowles	N/A	N/A
8059918	12/2010	Young	N/A	N/A
11652007	12/2022	Lin	378/70	G01N 23/083
2003/0094031	12/2002	Huang	N/A	N/A
2003/0094032	12/2002	Rodionovich et al.	N/A	N/A
2003/0158470	12/2002	Wolters et al.	N/A	N/A
2005/0069082	12/2004	Buijsse	N/A	N/A
2007/0047794	12/2006	Lang	N/A	N/A
2008/0219409	12/2007	Ueda	N/A	N/A
2009/0200604	12/2008	Chidambarao	N/A	N/A
2010/0008558	12/2009	Baeumer	N/A	N/A
2010/0147835	12/2009	Mulpuri	219/121.85	H01L 21/02491
2010/0296070	12/2009	Shibazaki	355/53	G03F 7/707
2010/0331782	12/2009	Hendriks	N/A	N/A
2011/0007869	12/2010	Gendreau	N/A	N/A
2011/0036360	12/2010	Lang	N/A	N/A
2011/0186914	12/2010	Cheng	257/E21.177	H01L 29/66772
2012/0219916	12/2011	Tashima	N/A	N/A
2013/0173214	12/2012	Yamashita	702/170	G01B 7/023
2014/0064445	12/2013	Adler	378/43	G01N 23/083
2014/0268122	12/2013	Matsumoto	N/A	N/A
2015/0036805	12/2014	Fu	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101261306	12/2007	CN	N/A
101459102	12/2008	CN	N/A
101996398	12/2010	CN	N/A
200506357	12/2004	TW	N/A
201310560	12/2012	TW	N/A
0028577	12/1999	WO	N/A

Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE (1) This application is a continuation application of U.S. patent application Ser. No. 16/666,196, filed Oct. 28, 2019, which is a continuation application of U.S. patent application Ser. No. 14/092,256, filed Nov. 27, 2013, now U.S. Pat. No. 10,460,999, issued Oct. 29, 2019, which is herein incorporated by reference in its entirety.

BACKGROUND

Technical Field

(1) The present disclosure relates to a metrology device.

Description of Related Art

(2) With the advancement of semiconductor technology, semiconductor devices become increasingly smaller and denser. The increasingly smaller and denser semiconductor devices are difficult to be manufactured and thus have inconsistent quality. Therefore, the semiconductor devices are examined before the semiconductor devices are released.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings are included to provide a further understanding of the disclosure, and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments of the disclosure and, together with the description, serve to explain the principles of the disclosure. In the drawings,

(2) FIG. 1 is a schematic diagram of a metrology device according to one or more embodiments of the present disclosure;

(3) FIG. 2 is a flow chart of a metrology method according to one or more embodiments of the present disclosure;

(4) FIG. 3 is a cross-sectional view of the wafer in FIG. 1;

(5) FIG. 4 is a 1D intensity graph of X-ray scattering spectrum measured from the wafer in FIG. 3; and

(6) FIG. 5 is a cross-sectional view of the wafer according to one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

(7) It is to be understood that the following disclosure provides many different embodiments, or examples, for implementing different features of the disclosure. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. Moreover, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed interposing the first and second features, such that the first and second features may not be in direct contact. Various features may be arbitrarily drawn in different scales for the sake of simplicity and clarity.

(8) The terms used in this specification generally have their ordinary meanings in the art and in the specific context where each term is used. The use of examples in this specification, including

examples of any terms discussed herein, is illustrative only, and in no way limits the scope and meaning of the disclosure or of any exemplified term. Likewise, the present disclosure is not limited to various embodiments given in this specification.

(9) As used herein, the terms “comprising,” “including,” “having,” “containing,” “involving,” and the like are to be understood to be open-ended, i.e., to mean including but not limited to. As used herein, “around”, “about”, “substantially” or “approximately” shall generally mean within 20 percent, in some embodiments within 10 percent, and in another embodiments within 5 percent of a given value or range. Numerical quantities given herein are approximate, meaning that the term “around”, “about”, “substantially” or “approximately” can be inferred if not expressly stated. The singular forms “a”, “an” and “the” used herein include plural referents unless the context clearly dictates otherwise.

(10) Reference throughout the specification to “one embodiment” or “an embodiment” means that a particular feature, structure, implementation, or characteristic described in connection with the embodiment is included in at least one embodiment of the present disclosure. Thus, uses of the phrases “in one embodiment” or “in an embodiment” in various places throughout the specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, implementation, or characteristics may be combined in any suitable manner in one or more embodiments.

(11) Reference is made to FIG. 1 which is a schematic diagram of a metrology device according to one or more embodiments of the present disclosure. The metrology device includes a light source **100** and an image sensor **200**. The light source **100** is configured for providing an X-ray **102** illuminating a wafer **900**. The image sensor **200** is configured for detecting a spatial domain pattern produced when the X-ray **102** illuminating the wafer **900**.

(12) As such, the metrology device of this embodiment can be utilized to measure smaller scale structures of the wafer **900** since the X-ray **102** has shorter wavelength, and the reliability of the measurement result can be improved.

(13) In one or more embodiments, the light source **100** may be an electromagnetic radiation source. More particular, the light source **100** can be an X-ray source. As a form of electromagnetic radiation, the X-ray **102** may have a wavelength in the range 0.01 to 10 nm corresponding to frequencies in the range 30 petahertz to 30 exahertz (3×10^{16} Hz to 3×10^{19} Hz). The wavelength of the X-ray **102** can be shorter than that of ultraviolet light. Since the optical resolution, which can be obtained from Rayleigh criterion, may be about half wavelength of the light beam illuminating to the wafer **900**, the optical resolution of the X-ray **102** may be about 0.005 to 5 nm. Taking the wafer **900** having linewidths which are about 20 nm as an example, the optical resolution of the X-ray **102** can be enough to distinguish the morphology of the wafer **900**.

(14) In this embodiment, the wafer **900** can be disposed between the light source **100** and the image sensor **200**. In other words, the image sensor **200** can detect the X-ray **102** passing through the wafer **900**. Therefore, compared to the reflective portion of the X-ray **102**, the transmission portion of the X-ray **102** can form a smaller spot size illuminating on the wafer **900**, and a smaller scale structure of the wafer **900** can be measured. In addition, although the X-ray **102** passes through the wafer **900** during measurement, the X-ray **102** can not damage the wafer **900**.

(15) Furthermore, the power of the X-ray **102** can be larger than 12 kW to pass through the wafer **900**. Taking the wafer **900** having a thickness 750 μm for example, the power of the X-ray **102** can be about 13 kW to 14 kW. However, in other embodiments, if the thickness of the wafer **900** has been reduced (such as being polished) before measurement, the power of the X-ray **102** can be smaller than 13 kW. Basically, the power of the X-ray **102** can be depend on the thickness of the wafer **900**.

(16) In FIG. 1, the metrology device can further include a lens group **300** for focusing the X-ray **102** to the wafer **900**. To measure the smaller scale structure of the wafer **900**, the spot size of the X-ray **102** illuminating on the wafer **900** can be smaller. Therefore, the spot size of the X-ray **102**

can be reduced by focusing the X-ray **102**. In one or more embodiments, the spot size of the X-ray **102** can be about 100 μm . It should be understood that the spot size of the X-ray **102** mentioned above is illustrative only and should not limit the scope of the claimed invention. A person having ordinary skill in the art may select a proper spot size of the X-ray **102** according to actual requirements.

(17) In this embodiment, the lens group **300** can be disposed between the light source **100** and the image sensor **200**. More specifically, the lens group **300** can be disposed between the light source **100** and the wafer **900**. As such, the X-ray **102** provided by the light source **100** can pass through the lens group **300**, being focused by the lens group **300**, and then illuminating to the wafer **900**. After passing through the wafer **900**, the transmission portion of the X-ray **102** can form the spatial domain pattern and then be detected by the image sensor **200**.

(18) In one or more embodiments, the metrology device can further include a processor **500** for obtaining the morphology of the wafer **900** according to the detected spatial domain pattern. The type of the processor **500** can depend on the type of the detected spatial domain pattern.

(19) In one or more embodiments, the metrology device may further include a Fourier's transformer **400** for transforming the detected spatial domain pattern into a Fourier frequency domain data, and the processor **500** can obtain the morphology of the wafer **900** according to the Fourier frequency domain data. In greater detail, interference may occur in the transmission portion of the X-ray **102** and form the spatial domain pattern. In other words, the spatial domain pattern can be an interference pattern. The Fourier's transformer **400** can transform the interference pattern into the Fourier frequency domain data, such that the processor **500** can obtain the morphology of the wafer **900** according to the Fourier frequency domain data.

(20) In this embodiment, the Fourier's transformer **400** can be connected to the image sensor **200**, and the processor **500** can be connected to the Fourier's transformer **400**. However, the claimed scope should be not limited to this respect. A person having ordinary skill in the art may design proper connections among the Fourier's transformer **400**, the processor **500**, and the image sensor **200** according to actual requirements.

(21) In one or more embodiments, the X-ray **102** may illuminate the wafer **900** from the back side of the wafer **900**. It should be pointed out that the back side can be a side opposite to a structure of the wafer **900** which is desired to be measured. After passing through the wafer **900**, diffraction may occur within the X-ray **102** due to the interference effect caused from the measured structure of the wafer **900**. This portion of the X-ray **102** may undergo destructive or constructive interferences, such that the interference pattern detected by the image sensor **200** may be not so easy to be analyzed in the spatial domain. However, the Fourier's transformer **400** can transfer the interference pattern into the frequency domain, such that some frequency features of the wafer **900** may be shown. The data corresponded to the frequency features can be analyzed using the processor **500**, for example, the processor **500** can transfer the data from the frequency domain back to the spatial domain, and the features of the wafer **900** can be obtained.

(22) In some examples, if the structure of the wafer **900** is periodic in the spatial domain, its frequency domain pattern may exist at least one main frequency which is corresponded to the period of the structure. That is, the main frequency is higher if the period of the structure is shorter, and the main frequency is lower if the period of the structure is longer. Therefore, according to the main frequency, the feature of the wafer **900** can be obtained.

(23) In one or more embodiments, the metrology device may further include a rotation mechanism **600** for rotating the wafer **900**. The rotating mechanism **600** can be disposed between the lens group **300** and the image sensor **200**. However, the claimed scope is not limited to this respect. Through rotating the wafer **900** during measurement, more information of the wafer **900** can be obtained. In greater detail, the X-ray **102** may interact with the electrons and then scatter when it illuminates the wafer **900**. If the structure of the wafer **900** is uniform and disordered, the scattered X-ray **102** may be uniformly scattered in the spatial domain. However, if the electron distribution

or the structure of the wafer **900** is ordered or regular, the feature of the scattered X-ray **102** may be changed according to the different incident angle of the X-ray **102**. Therefore, for rotating the wafer **900** during measurement, the directional features of the wafer **900** can be obtained.

(24) It is understood that the embodiment of the metrology device mentioned above is provided merely as examples and are not intended to be limiting. The metrology device may have different configurations consistent with the spirit of the present disclosure in alternative embodiments depending on design requirements and manufacturing concerns.

(25) Another aspect of the present disclosure provides a metrology method for measuring the morphology of a wafer. Reference is made to FIG. 2 which is a flow chart of the metrology method according to one or more embodiments of the present disclosure. To illustrate clarity, the structure of the metrology device in FIG. 1 is applied to the method shown in FIG. 2 in this embodiment. However, the claimed scope is not limited to this respect. The metrology method for measuring the morphology of the wafer **900** includes the following acts:

(26) As shown in act S10, the X-ray **102** is provided to illuminate the wafer **900**. Subsequently, as shown in act S20, the spatial domain pattern produced when the X-ray **102** illuminating the wafer **900** is detected. It should be noticed that the flow chart of FIG. 2 shows exemplary acts, but they are not necessarily performed in the order shown. Acts may be added, replaced, changed order, and/or eliminated as appropriate, in accordance with the spirit and scope of disclosed embodiments.

(27) In greater detail, the wavelength of the X-ray **102** can be about 0.01 to 10 nm, such that the optical resolution of the X-ray **102**, which may be about 0.005 to 5 nm, can be further improved.

(28) In one or more embodiments, the metrology method may further include act: focusing the X-ray **102** to the wafer **900**.

(29) For example, the spot size illuminating on the wafer **900** may be 100 μm . Therefore, the spot size of the X-ray **102** can be reduced to measure the smaller scale structures of the wafer **900** if the wafer **900** has smaller devices.

(30) The following paragraphs provide an example with respect to the data of measuring the wafer **900** using the metrology device of FIG. 1. FIG. 3 is a cross-sectional view of the wafer **900** in FIG. 1, and FIG. 4 is a 1D intensity graph of X-ray scattering spectrum measured from the wafer **900** in FIG. 3. Reference is made to FIG. 3 and FIG. 4. In this example, a measured structure **902** of the wafer **900** was a metal layer structure. The thickness T of the wafer **900** was about 750 nm. The wavelength of the X-ray was about 0.1 nm. The power of the X-ray was about 14 kW. The X-ray was incident the back side of the wafer **900** normally, i.e., the incident angle of the X-ray was 0 degrees.

(31) In FIG. 4, the detected spatial domain pattern was the 2D X-ray scattering spectrum (not shown), and the intensity of the 2D X-ray scattering spectrum along x direction was plotted as the “Experimental” curve in FIG. 4. Subsequently, the morphology was obtained. In this example, the “Experimental” curve was fitted to obtain a fitting curve, which was shown in the “Fitting” curve in FIG. 4. According to the “Fitting” curve, main peaks (repeat in scattering vector $Q_x \sim 0.0083 \text{ \AA}^{-1}$) are obtained. According to the scattering vector, pitch $P = 75.79 \text{ nm}$ was estimated. In addition, average width $W = 30.46 \text{ nm}$ of the measured structure **902** was also estimated. In this example, the values of the pitch P and the average width W of the wafer **900** was smaller than the optical resolution of ultraviolet light, which has wavelength about 150 nm and optical resolution about 75 nm. As such, since the optical resolution of the metrology device in this example was much higher, the metrology device can achieve higher measurement reliability.

(32) In other embodiments, the wafer **900** in FIG. 3 can further be rotated during the measurement, such that more information of the morphology of the wafer **900** can be obtained. For example, top width TW of the measured structure **902**, middle width MW of the measured structure **902**, and the bottom width BW of the measured structure **902** can be obtained. It should be understood that the feature of the wafer **900** mentioned above is illustrative only and should not limit the scope of the claimed disclosure. A person having ordinary skill in the art may select a proper feature of the

wafer **900** to be measured according to actual requirements.

(33) It should be noticed that in other embodiments, the measured structure **902** of the wafer **900** is not limited to the structure mentioned above. According to different manufacturing processes, the wafer **900** in FIG. **3** may be a Fin field effect transistor (FinFET), and in other embodiments, the wafer **900** in FIG. **3** may include conductive layers, dielectric layers, and/or semiconductor layers, such that the measured structure can be the morphology of the metal layer of the wafer **900**, the morphology of the dielectric layer of the wafer **900**, or the morphology of semiconductor layers of the wafer **900**.

(34) In greater detail, the conductive layers may include metal layers (the material of which is such as Ag, Au, Cu, etc.), or transparent conductive oxide layers (the material of which is such as ITO, IZO, etc.). The dielectric layer may include oxide layers (such as SiO.sub.x), or nitride layers (the material of which is such as SiN_x). The semiconductor layers may include silicon layers (the material of which is such as poly-Si, a-Si, etc.). However, the claimed scope should not be limited to the layers mentioned above.

(35) Reference is made to FIG. **5** which is a cross-sectional view of the wafer **900** according to one or more embodiments of the present disclosure. In this embodiment, the wafer **900** can further include raised source and drain structures **904**. The raised source and drain structures **904** can be grown on or above a substrate **910** of the wafer **900** using the epi-growth technique. The metrology device in this embodiment can measure the raised height H, pitches P, and/or widths W of the raised source and drain structures **904**. Furthermore, through rotating the wafer **900**, more information of the morphology of the wafer **900** can be obtained.

(36) According to some embodiments, a method includes illuminating a wafer by an X-ray, detecting a spatial domain pattern produced when illuminating the wafer by the X-ray, identifying at least one peak from the detected spatial domain pattern, and analyzing the at least one peak to obtain a morphology of a transistor structure of the wafer.

(37) According to some embodiments, a method includes illuminating a wafer by an X-ray, such that the X-ray is diffracted by the wafer, detecting the diffracted X-ray, conducting a Fourier's transform of the diffracted X-ray, and analyzing a main frequency of the Fourier's transform of the diffracted X-ray to obtain a morphology of a transistor structure of the wafer.

(38) According to some embodiments, a method includes illuminating a wafer by an X-ray, detecting an X-ray scattering spectrum produced when illuminating the wafer by the X-ray, obtaining an experimental curve of intensity of the X-ray scattering spectrum as a function of scattering vector, and analyzing the experimental curve to obtain an average width of measured structures of the wafer.

(39) The acts are not recited in the sequence in which the acts are performed. That is, unless the sequence of the acts is expressly indicated, the sequence of the acts is interchangeable, and all or part of the acts may be simultaneously, partially simultaneously, or sequentially performed.

(40) Although the embodiments have been described in detail, it should be understood that various changes, substitutions and alterations can be made herein without departing from the spirit and scope of the embodiments as defined by the appended claims. For example, the wavelength of the X-ray **102** is not limited to the wavelength range mentioned above.

(41) Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, and composition of matter, means, methods, and steps described in the specification. A person having ordinary skill in the art can readily appreciate from the disclosure, processes, machines, manufacture, compositions of matter, means, methods, or steps, presently existing or later to be developed, that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein may be utilized according to the disclosure. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods, or steps. In addition, each claim constitutes a separate embodiment, and the

combination of various claims and embodiments are within the scope of the disclosure. Accordingly, the scope of the present disclosure is defined by the appended claims rather than the foregoing description and the exemplary embodiments described therein.

Claims

1. A method, comprising: placing a wafer on a rotation mechanism of a metrology device; emitting an X-ray from the metrology device onto a backside of the wafer such that the X-ray converges to a point on the backside of the wafer, wherein a power of the X-ray is greater than about 12 kW; rotating, by using the rotation mechanism, the wafer while illuminating the wafer by the X-ray; detecting, by using an image sensor of the metrology device, a transmission portion of the X-ray passing through the wafer; and obtaining, by using a processor of the metrology device, a top width and a bottom width of a structure over the wafer based on the transmission portion of the X-ray with different rotating angles of the rotation mechanism.
2. The method of claim 1, wherein the structure is a metal layer structure.
3. The method of claim 1, wherein the structure is periodic in a spatial domain.
4. The method of claim 1, wherein the structure is over a front side of the wafer that is opposite the backside of the wafer.
5. The method of claim 1, wherein obtaining the top width and the bottom width of the structure comprises: conducting a Fourier's transform of the detected transmission portion of the X-ray; and obtaining the top width and the bottom width from the Fourier's transform of the detected transmission portion of the X-ray.
6. The method of claim 1, wherein the structure comprises a FinFET transistor.
7. The method of claim 1, further comprising: polishing the wafer prior to placing the wafer on the rotation mechanism of the metrology device.
8. The method of claim 1, wherein a spot size the point of the X-ray on the backside of the wafer is about 100 μm .
9. A method, comprising: polishing a wafer; after polishing the wafer, placing the wafer between a light source and an image sensor of a metrology device, wherein a backside of the wafer faces the light source; illuminating, by using the light source, the wafer with an X-ray, wherein a power of the X-ray is determined by a thickness of the wafer; rotating the wafer along a diameter axis of the wafer; sensing, by using the image sensor, the X-ray after the X-ray passes through the wafer during rotating the wafer; and analyzing, by using a processor of the of metrology device, the X-ray to obtain a morphology of a transistor structure of the wafer.
10. The method of claim 9, wherein the power of the X-ray is about 13 kW to about 14 kW.
11. The method of claim 9, wherein the transistor structure of the wafer comprises a raised source/drain structure, and the morphology of the transistor structure is a height of the raised source/drain structure.
12. The method of claim 9, wherein the transistor structure of the wafer comprises a raised source/drain structure, and the morphology of the transistor structure is a width of the raised source/drain structure.
13. The method of claim 9, wherein the transistor structure of the wafer comprises a first raised source/drain structure and a second raised source/drain structure, and the morphology of the transistor structure is a distance between the first raised source/drain structure and the second raised source/drain structure.
14. The method of claim 11, wherein the raised source/drain structure is an epitaxial structure.
15. The method of claim 9, wherein a beam of the X-ray progressively narrows until the X-ray converges to a single point on the backside of the wafer.
16. A method, comprising: providing, by using a light source of a metrology device, an X-ray; focusing, by using a lens group of the metrology device, the X-ray onto a backside of a wafer, a

beam of the X-ray gradually narrowing until the X-ray converges to a single point on the backside of the wafer, wherein a power of the X-ray is greater than about 12 kW; detecting, by using an image sensor of the metrology device, a diffracted portion of the X-ray; conducting, by using a Fourier's transformer of the metrology device, a Fourier's transform of the diffracted portion of the X-ray; identifying, by using a processor of the metrology device, at least one peak from the Fourier's transform; and analyzing, by using the processor, the at least one peak to obtain a height of a source/drain feature of a transistor structure of the wafer.

17. The method of claim 16, wherein conducting the Fourier's transform of the diffracted portion of the X-ray is performed to obtain an experimental curve as a function of a scattering vector, and identifying at least one peak from the Fourier's transform comprises: choosing a fitting curve, wherein a main peak of the experimental curve and a main peak of the fitting curve are at substantially the same scattering vector, and the main peak of the experimental curve is the at least one peak.

18. The method of claim 16, wherein a spot size of the single point of the X-ray focusing onto the backside of the wafer is about 100 μm .

19. The method of claim 16, further comprising: polishing the wafer prior to focusing the X-ray onto the backside of the wafer.

20. The method of claim 16, wherein analyzing the at least one peak is further performed to obtain a morphology of a dielectric layer of the wafer.
